Thakur Educational Trust's (Regd.)
THAKUR SHREE DPS COLLEGE OF ENGINEERING & MANAGEMENT
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Blueprint for Applied physics

Department: F.E. (All Branches)

Internal Assessment Test-II (A.Y. 2025-26)

Subject: Applied Physics **NEP-2020** **Sem: I**

Max. Marks: 15 **Duration: 1 Hour**

Note: All questions are compulsory.

Q. No.	Questions	Marks	BL	CO
Q.1	**Attempt any two of the following:**			
1(a)	Theory of semiconductors based on energy band gap; intrinsic and extrinsic semiconductors.	3	L2	CO6
1(b)	Numerical problem on conductivity and mobility/ Fermi-Dirac distribution function.	3	L3	CO6
1(c)	Theory and derivation of the Fermi level in intrinsic and extrinsic semiconductors.	3	L2	CO6
Q.2	**Attempt any one of the following:**			
2(a)	Numerical problems on gradient, divergence and curl, along with their physical significance.	4	L3	CO4
2(b)	Derivation of Maxwell's equations in free space.	4	L3	CO4
Q.3	**Attempt any one of the following:**			
3(a)	Theory and numerical problem based on the de Broglie hypothesis, Heisenberg's uncertainty principle and its application.	5	L2, L3	CO5
3(b)	Derivation of Schrödinger's time-dependent and independent wave equations.	5	L2, L3	CO5

BL: Bloom's Taxonomy Levels(L1: Remembering, L2: Understanding, L3: Applying, L4: Analyzing, L5: Evaluating, L6: Creating)
CO: Course Outcomes